

# MTH205 Analysis in one variable

## Tutorial 1

Date: **10.01.2025**

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- (1) Let  $x \in (-1, \infty)$ . Prove that  $(1+x)^n \geq 1+nx$  for every  $n \in \mathbb{N}$ .
- (2) Let  $X$  be a non-empty set and  $\mathbb{P}(X)$  denote the power set of  $X$ . Show that there does NOT exist any surjective map  $f: X \rightarrow \mathbb{P}(X)$ .
- (3) Prove the following:
- (a)  $(0, 1] = \bigcap_{n=1}^{\infty} (0, 1 + \frac{1}{n})$ .
  - (b) Let  $a, b \in \mathbb{R}$ . Then the intervals  $(a, \infty)$  &  $(-\infty, a)$  are open sets and the intervals  $[b, \infty)$  &  $(-\infty, b]$  are closed sets.
  - (c)  $\mathbb{N}$ , the set of natural numbers is a closed set.
  - (d)  $\mathbb{Q}$ , the set of rational numbers is neither open nor closed.
  - (e) If  $G$  is an open set and  $F$  is a closed set in  $\mathbb{R}$  then  $G \setminus F$  is an open set and  $F \setminus G$  is a closed set.
- (4) Let  $[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \dots$  be a decreasing sequence of non-empty closed intervals in  $\mathbb{R}$ . Then show that  $\bigcap_{n=1}^{\infty} [a_n, b_n] \neq \emptyset$ . If in addition  $(b_n - a_n) \rightarrow 0$  as  $n \rightarrow \infty$  then what about the intersection?
- (5) A subset  $G$  of  $\mathbb{R}$  is open if and only if  $G$  is a countable union of disjoint open intervals.
- (6) Show that the Cantor set is uncountable.